

NYCU-ECE DCS-2026

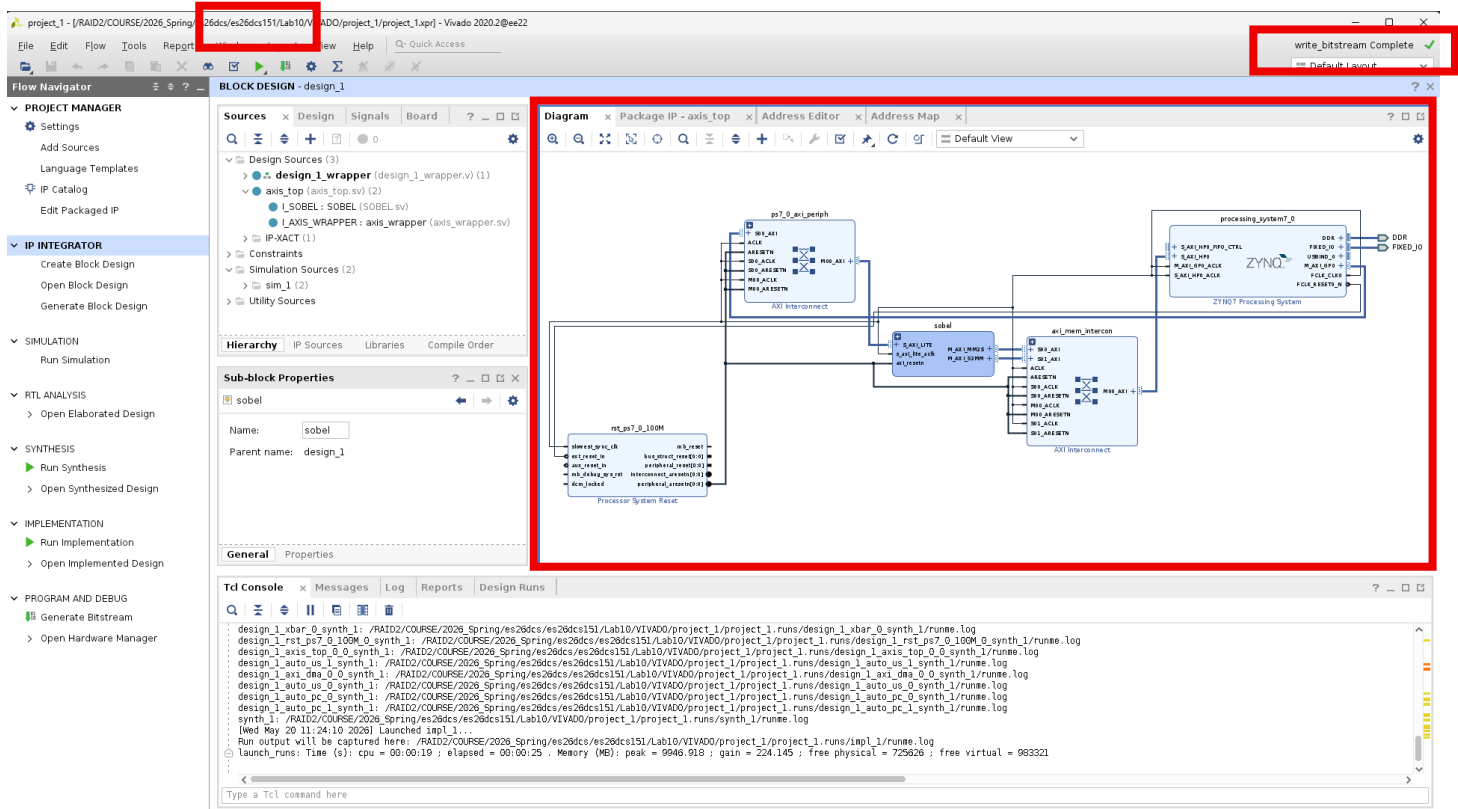
Lab10 Report

Name: 楊庭瑞

Student ID: 113511266

Server Account: dcs151

1. Vivado 截圖 (帳號, write bitstream Complete, Block diagram)



2. Jupyter Notebook (-3. 執行軟體與硬體運算)

```
In [6]: # =====
# 3. 執行軟體與硬體運算
# =====
print(f"=== es26dcs(workstation_ID) Result ===\n")
print("--- Software Execution (Python) ---")
sw_start = time.time()
sw_results = software_sobel(test_input_data)
sw_exec_time = time.time() - sw_start
print(f"execution time: {sw_exec_time:.6f} s\n")

print("--- Hardware Execution (FPGA) ---")
hw_start = time.time()

# 啟動 DMA 傳輸與接收
dma.sendchannel.transfer(in_buffer)
dma.sendchannel.wait()
dma.recvchannel.transfer(out_buffer)
dma.recvchannel.wait()

hw_exec_time = time.time() - hw_start
print(f"execution time: {hw_exec_time:.6f} s\n")

# 解析硬體結果
hw_results = [int(val) & 0x1 for val in out_buffer]

=== es26dcs151 Result ===

--- Software Execution (Python) ---
execution time: 0.010025 s

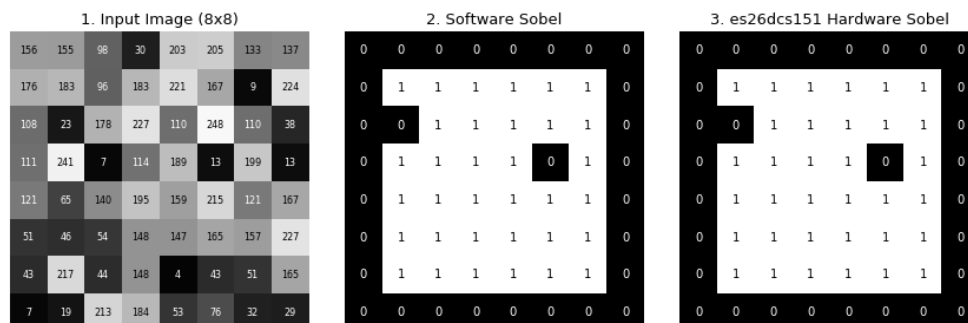
--- Hardware Execution (FPGA) ---
execution time: 0.002115 s
```

3. Jupyter Notebook (-4. 效能報告與視覺化比對)

=== es26dcs151 Report ===

✅ 驗證成功，軟體與硬體運算結果完全一致

🚀 Speedup: 4.74 x



4. Jupyter Notebook (-8. 顯示圖片成果)

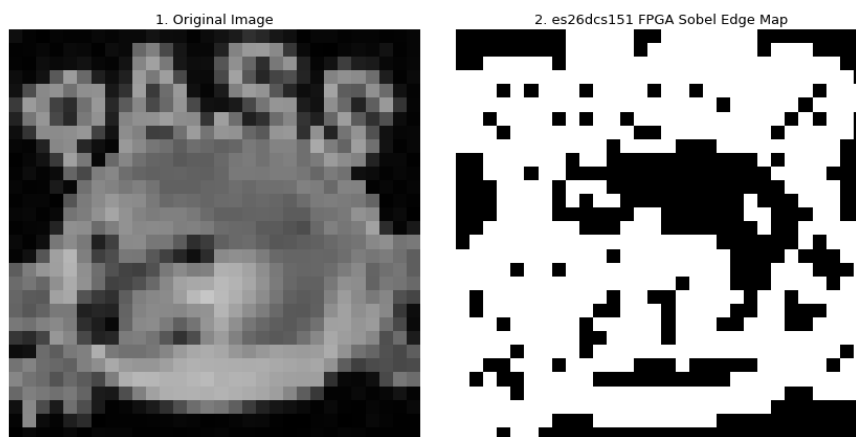
```
In [11]: # =====
# 8. 顯示圖片成果
# =====
print(f'=== es26dcs{workstation_ID} Image Result ===')
fig, axes = plt.subplots(1, 2, figsize=(12, 6))

plot_gray_panel(axes[0], img_array, "1. Original Image", vmin=0, vmax=255)
plot_gray_panel(axes[1], final_edge_map, f"2. es26dcs{workstation_ID} FPGA Sobel Edge Map", vmin=0, vmax=1)

plt.tight_layout()
plt.show()

# 釋放記憶體
del in_buffer, out_buffer

=== es26dcs151 Image Result ===
```



想對助教說的話 or 心得:

這次 Lab 是本課程的最後一次，很開心自己撐完了整個學期，也感謝出這次 Lab 的助教這麼用心的在 FB 回答大家的問題，簡報的每個步驟說明很清楚，每個步驟截圖截的很細，辛苦助教了！